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Development and interface characterization of unmatched glass-metal joint

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Abstract

An unmatched glass-metal joint between borosilicate glass and austenitic stainless steel SS 304 was developed for parabolic trough receiver tube application. $40\text{SiO}_2\text{-}54\text{B}_2\text{O}_3\text{-}4\text{Na}_2\text{O-}2\text{Al}_2\text{O}_3$ material was used as the glass sealant to join the SS304 metal ring and borosilicate glass tube. The research work reported was carried out with objective of investigating the role of glass sealant, joint design and pre-oxidation conditions on the wettability and adhesion strength of the glass-metal joint. The work also aimed at analysing the types of oxides formed at the glass-metal joint interface. The pre-oxidation behavior of SS304 was studied by varying the pre-oxidation temperature and pre-oxidation time. The oxide formed on the surface of SS304 substrates were examined using x-ray diffraction and energy dispersive x-ray spectroscopy. During the glass-metal sealing process the oxides at the glass-metal interface were characterized using x-ray diffraction

Outline

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and scanning electron microscopy. In-situ wetting behavior of glass sealant $40\text{SiO}_2\text{-}54\text{B}_2\text{O}_3\text{-}4\text{Na}_2\text{O-}2\text{Al}_2\text{O}_3$ over austenitic stainless steel SS 304 was examined using the in-house developed image acquisition system. The optimum condition for joining $40\text{SiO}_2\text{-}54\text{B}_2\text{O}_3\text{-}4\text{Na}_2\text{O-}2\text{Al}_2\text{O}_3$ glass sealant with SS304 is obtained by pre-oxidizing metal specimen at 700°C for 3 min, where optimum oxide layer mass gain of $5.8666 \times 10^{-4} \text{g/cm}^2$ and optimum oxide layer thickness of $87.57 \mu\text{m}$ was achieved. The maximum strength of joint during adhesion test was measured around 2.57N/mm^2 (MPa) corresponding to pre-oxidation temperature of 700°C and 3 min pre-oxidation time. The FeO and $((\text{SiO}_2), \text{AlNa}(\text{SiO}_4))$ oxide phases were observed at the interface of glass and the metal during joining. During in-situ wettability analysis the contact angle between the glass sealant pallet and SS304 substrate was measured at different temperatures. The contact angle of the glass sealant pallet decreased during heating process and the contact angle increased as the glass sealant pallet cooled from sealing temperature.

Keywords

Glass-metal joint; Parabolic trough receiver tube; Pre-oxidation; XRD; SEM; Induction heating; Wettability

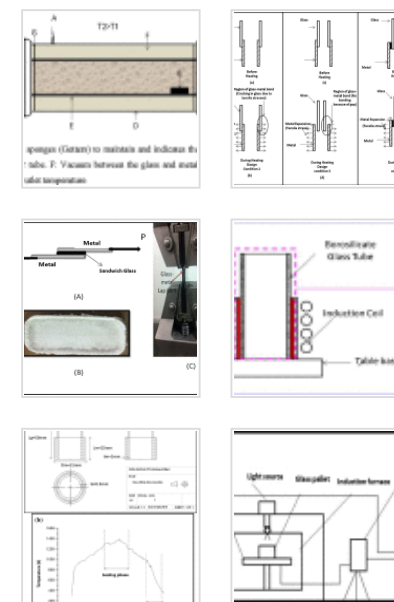
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1. Introduction

The joining of glass with metal remains a challenge because of the amorphous and crystalline nature of glass and metal respectively. Glass to metal joint has been used for many applications such as vacuum tube, parabolic trough receiver tube, solid oxide fuel cell, and electronic semiconductor applications [1,2]. The glass metal joint can be broadly classified into two categories, viz., matched glass metal joint (the coefficient of thermal expansion is same or closely matched for glass and metal) unmatched glass metal joint

Figures (19)



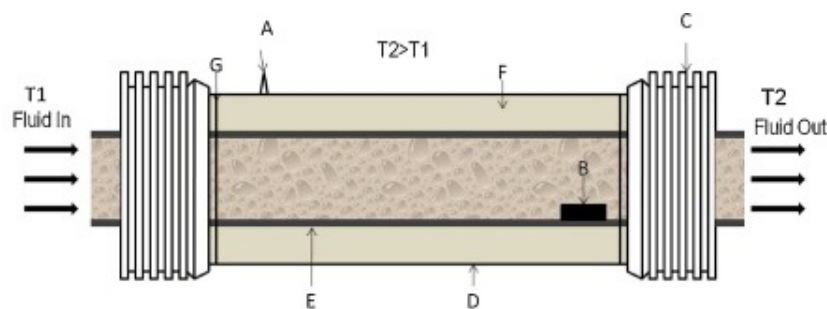
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(coefficient of thermal expansion is significantly different for glass and the metal). Glass-metal joining is traditionally a fusion technique, in which glass is melted over the metal surface and reacts to form an interface. The bonding of glass with the metal mainly depends upon the type of oxide present at the metal surface after pre-oxidation. The oxide formed on the metal surface after pre-oxidation acts as a bonding agent at the interface between the glass and the metal. The properties and the types of oxides formed at the interface affect the bonding of the glass with the metal [1].

It is also recognized that the residual stresses generated due to difference in coefficient of thermal expansion of the glass and the metal must be kept as low as possible [2]. The generated thermal residual stress at glass-metal interface breaks the glass tube at the interface [3], [4], [5], [6]]. The breaking of the glass at the glass-metal interface leads to the loss of vacuum in the parabolic trough receiver tube. A Parabolic Trough Receiver Tube (PTRT) consists of steel pipe surrounded by transparent glass tube and the space between them consisting of vacuum. The steel pipe is coated with a solar selective coating which is responsible for absorbing maximum solar radiation and minimizing reflection and radiation losses as suggested by Barriga et al. [7]. The working fluid temperature is around 265 °C as reported in the literature [8,9]. The geometry of a parabolic trough receiver tube is shown in Fig. 1.



A: Evacuation Tube B: Chemical sponges (Getters) to maintain and indicates the status of vacuum. C: Bellow. D: Glass envelops. E: Steel absorber tube. F: Vacuum between the glass and metal tube. G: Glass to metal joint. T_1 : Fluid inlet temperature. T_2 : Fluid outlet temperature

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Fig. 1. The geometry of parabolic trough receiver tube.

A: Evacuation Tube B: Chemical sponges (Getters) to maintain and indicates the status of vacuum. C: Bellow. D: Glass envelops. E: Steel absorber tube. F: Vacuum between the glass and metal tube. G: Glass to metal joint. T_1 : Fluid inlet temperature. T_2 : Fluid outlet temperature.

From the literature [[4], [5], [6],10,[11], [12], [13]], it has been noted that kovar alloy (ASTM F-15 alloy) having a coefficient of thermal expansion $5.0 \times 10^{-6} \text{ m/m}^\circ\text{C}$ and kodial glass with a coefficient of thermal expansion $3.3 \times 10^{-6} \text{ m/m}^\circ\text{C}$ has been used for the development of a matched glass-metal joint for parabolic trough receiver tube [10]. The pre-oxidation behavior of kovar alloy has been studied by varying the pre-oxidation time and the pre-oxidation temperature along with different environment by Cheng-hsien and Pi-ying Cheng [11]. Wettability study of different glasses over kovar alloy has been studied using image capturing system by Chenghsien Kuo et al. [12]. The experimental results indicate that the pre-oxidation of kovar alloy for about 10 min at 700°C provides better adherence of glass with kovar [12]. FeO and Fe_3O_4 types of oxides were formed after pre-oxidation of kovar alloy as reported by D.W. Luo and Z.S. Shen [10].

Studies on glass-metal joining for parabolic trough receiver tube have been conducted using kovar alloy and kodial glass by researchers [[4], [5], [6]]. Wettability study of glass over kovar alloy has also been carried out at joining temperature of 980°C by Chenghsien Kuo et al. [12].

The aim of the present work was to fabricate unmatched (coefficient of thermal expansion of both glass and metal differ significantly) glass-metal joint between SS304 alloy having coefficient of thermal expansion ($17.3 \times 10^{-6} \text{ m/m}^\circ\text{C}$) and borosilicate glass with coefficient of thermal expansion ($3.3 \times 10^{-6} \text{ m/m}^\circ\text{C}$). The selection of SS 304 alloy and the borosilicate glass for the present study is based on their ease of availability and their lesser cost as compared to the kovar alloy and the kodial glass. Glass sealant was used to

fill the space between borosilicate glass tube and the SS304 metal ring. Glass sealant composition was selected on the basis of the shear strength tests. Localized heating was used through induction heating process to the join glass-metal assembly. Pre-oxidation behavior of the austenitic stainless steel SS304 alloy specimen was studied by varying pre-oxidation temperature and pre-oxidation time. The types of oxides formed after pre-oxidation were examined using x-ray diffraction (XRD) technique. The morphology of the glass-metal joint interface was analyzed with the help of scanning electron microscope and the X-ray dot mapping technique. In situ wetting behavior of glass sealant over the austenitic stainless steel alloy SS304 was analysed at different temperatures by using an image capturing system. The adhesive strength of the glass with the metal was measured using specially designed specimen holding fixture used for testing under compression loading condition.

2. Experimental detail

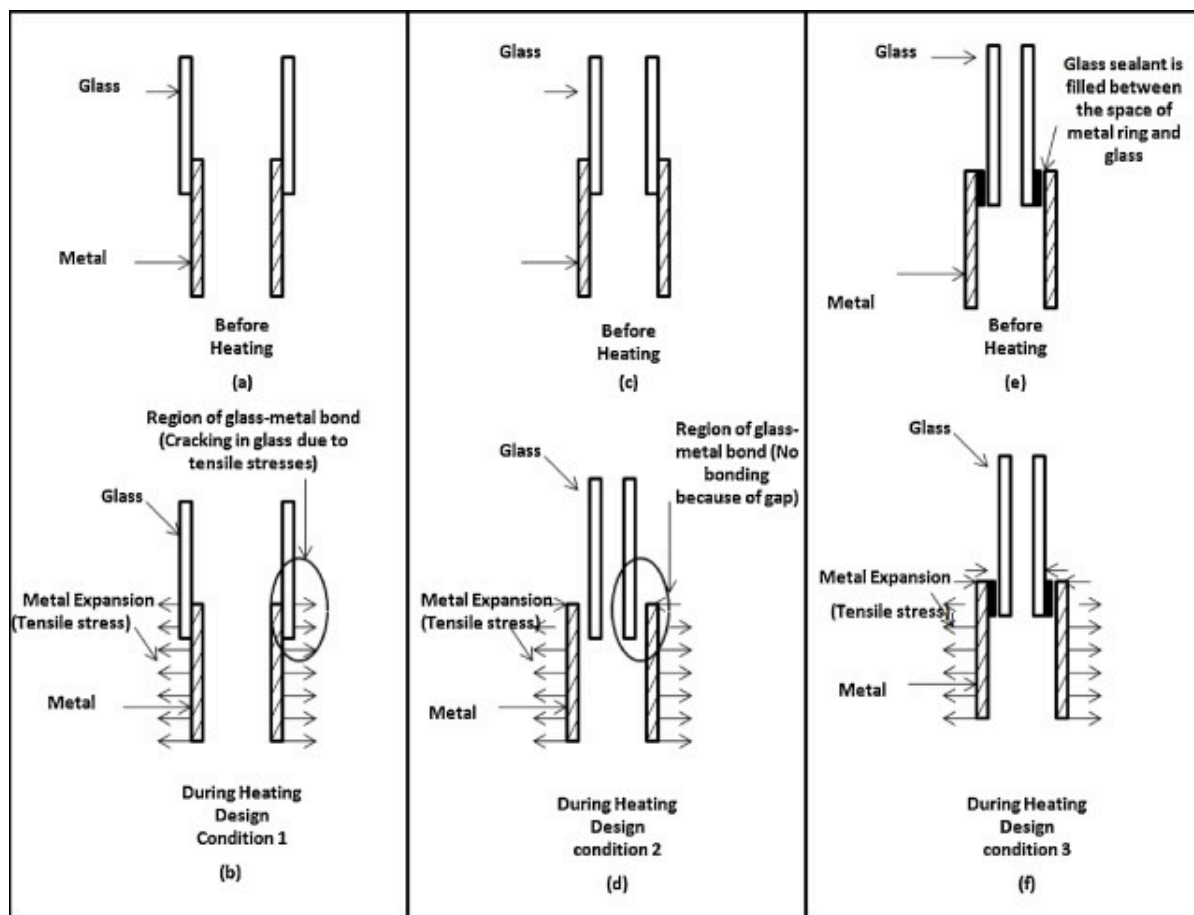
2.1. Material selection

In the past, researchers [[4], [5], [6]] have used kovar alloy and kodial glass combination for glass-metal sealing application because of their closely matched coefficient of thermal expansion. In the present work borosilicate glass tube and SS304 metal rings were joined. The major issue involving the development of glass metal joint is the formation of suitable bonding between the glass sealant and the glass tube on one side and glass sealant and metal ring on the other side. The problem is complicated due to the differential expansion of the glass sealant and the metal ring during heating due to a significant gap is formed between the metal ring and the glass sealant. This gap closes during cooling as metal ring shrinks much more as compared to glass sealant when cooled to room temperature. The gap between the metal ring and glass tube (where the glass sealant is filled) depends upon the initial design selection (geometric dimensions) of the metal ring and the glass tube dimensions. The larger initial gap would result in debonding of glass sealant with the metal ring as the metal ring would expand much

more in heated condition forcing the glass sealant to leak out. The smaller initial gap would limit the strength of the joint and can cause breakage of the glass tube. Hence the selection of the suitable glass tube–metal ring joint design is very important issue to obtain a suitable compression joint. The selection of the suitable glass sealant is also a major issue as the coefficient of thermal expansion mismatch between the glass sealant and the metal ring helps in increasing the development of compressive stresses on the glass sealant and prevents the breakage of glass tube due to absence of any direct compressive stresses on the glass tube. It is necessary to have a glass sealant with lower melting point than glass tube and have high wettability with the metal ring. The compressive strength of glass is larger than its tensile strength therefore compressive stresses generated on glass tube and glass sealant after fabrication of glass-metal joint helps to resist high mechanical stresses in glass-metal joint.

2.1.1. Selection of glass-metal joint design

The experimental procedure for adopting a suitable glass-metal joint design is shown in [Fig. 2](#). Initially the glass tube inner surface was placed over the outer surface of metal ring as shown in [Fig. 2\(a\)](#). The complete assembly was then heated locally to 980 °C temperature with the help of an induction heating set-up.



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Fig. 2. Different design condition of glass-metal joint.

During heating, it was observed that the metal expansion was faster than glass and resulted in breakage of glass as shown in Fig. 2(b). In other trial, the glass tube was placed in tight fit over the inner surface of metal tubes Fig. 2(c). The complete assembly was then heated to 980°C from room temperature in induction heating set-up. It was observed that metal expands faster than glass in the outward direction and a gap was formed between the metal ring and glass tube as presented in Fig. 2(d).

To overcome the problem originating from the design condition shown in Fig. 2(c) and (d), the suitable glass sealant was used to fill the gap between the metal ring and glass tube as shown in Fig. 2(e). During the heating of assembly from room temperature to joining temperature, the metal expands faster than the glass. The glass sealant started expanding in outward direction along with metal. However when the complete assembly was cooled to room temperature, the metal contracts to original position and compressive stresses were developed in glass as shown in Fig. 2(f). As a result of this experimentation, the compressive joint was formed between glass tube and metal ring with the help of the glass sealant. Researchers [[18], [19], [20]] have recommended the use of the compression joint in case there is a significant difference in coefficient of thermal expansion between the joining materials.

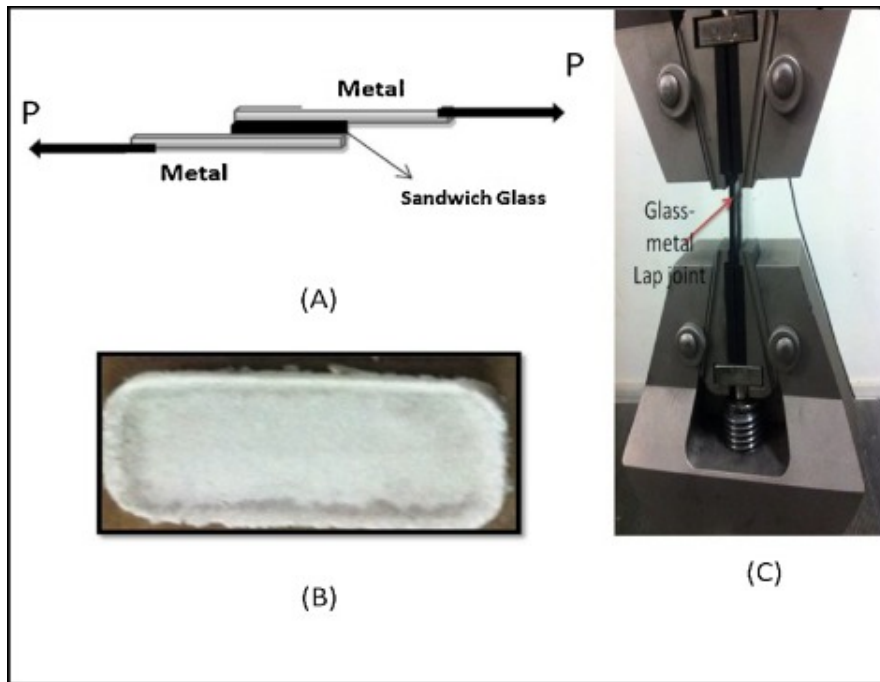
2.1.2. Selection of glass sealant

For the development of compression glass-metal joint it is necessary that the glass sealant should have a lower melting point as compared to that of the glass tube. Glass sealant should have high wettability in order to achieve a good level of bonding. To achieve high wettability in glass sealant, the percentage compositions of SiO_2 was decreased and percentage composition of B_2O_3 was increased while keeping the percentage composition of Al_2O_3 and Na_2O the same as shown in Table 1. The compression joint design was obtained with this combination. The dimension of compression glass-metal joining assembly used in present study is shown in Fig.5(a). Different composition glass sealants were prepared by mixing different compounds as shown in Table 1. Glass powder was ground using mortar and pestle after weighing and mixing in appropriate percentage. Around 3 wt.% polyvinyl alcohol (PVA) powder was mixed in cold water to prepare a viscous liquid which was used as glass binder to bind the glass powder. The glass powder was then pressed in die to form rectangular pallets of dimension $25.4 \text{ mm} \times 62.5 \text{ mm} \times 1.5 \text{ mm}$ as shown in Fig.3(b). Rectangular pallet was then baked in oven for 120°C in order to remove binder from glass through the process of evaporation. The glass sealant pallet was then placed between two pre-oxidized metal substrates having dimension $(25.4 \text{ mm} \times 25.4 \text{ mm} \times 1.5 \text{ mm})$. The size of metal plate and

glass sealant were selected in accordance with the (ISO) 4587 standard as used by Cheng-hsien and Pi-ying Cheng [11].

Table 1. Chemical composition and shear strength of different glass sealants [15].

Glass code	SiO ₂ (Wt %)	Al ₂ O ₃ (Wt %)	B ₂ O ₃ (Wt %)	Na ₂ O (Wt%)	Shear Strength (MPa)
G1	81	2	13	4	0
G2	75	2	19	4	0
G3	70	2	24	4	0
G4	60	2	34	4	0
G5	50	2	44	4	0
G6	40	2	54	4	0.12754



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Fig. 3. (a) Lap joint geometry. (b) Shape of glass sealant pallet. (c) Specimen mounted on universal tensile testing machine.

The shear strengths of glass sealants obtained are listed in [Table 1](#).

The complete assembly was heated to 980 °C for 20 min and then it was cooled within the furnace till room temperature. The shear strength of the fabricated glass-metal joint was tested with the universal tensile testing machine as shown in [Fig. 3\(c\)](#).

2.2. Pre-oxidation of alloy steel 304 specimen

The purpose of pre-oxidation was to form oxide film on the surface of SS304 alloy specimen. The strength of glass to SS304 alloy joint depends upon the pre-oxidation of the alloy. The quality of bond in glass-metal joint depends upon the oxide layer thickness

and the types of oxide formed at the alloy surface. The factors affecting the oxide layer thickness are pre oxidation time and pre oxidation temperature. To study the oxide layer formation on SS304 alloy specimen, 48 (SS304) metal specimens of dimension 25 mm × 25 mm × 1.5 mm were repeatedly heated in a muffle furnace at 700 °C for 2–5 min. The initial weights of these metal specimens before heating were measured and recorded using electronic precision weighing balance. After cooling to room temperature, the specimens were again weighed using a precision weighing balance with an accuracy of ±0.1 mg. The oxide layer thickness formed on metal specimen corresponding to different pre-oxidation conditions was measured using an optical microscope “Dewinter”. The pre-oxidized metal specimens were cut along the thickness of metal specimen to dimension of 10 mm × 10 mm square area and 1.5 mm thickness and then mounted in resin filled specimen holder.

2.3. Types of oxide formed after pre-oxidation

In order to analyze the type of oxides formed after pre-oxidation, pre-oxidized metal specimens were analyzed using “Bruker Axis Automated Powder X- Ray Diffractometer” with Cu K α ($\lambda = 1.54 \text{ \AA}$) radiation. During the experimentation, the scan speed was 0.02°/min. The elemental composition of metal oxide formed after pre-oxidation was measured with the help of “Energy Dispersive X ray Spectroscopy” EDAX. The morphology of oxide present at the metal surface was analysed using “Zeiss make Scanning Electron Microscope”.

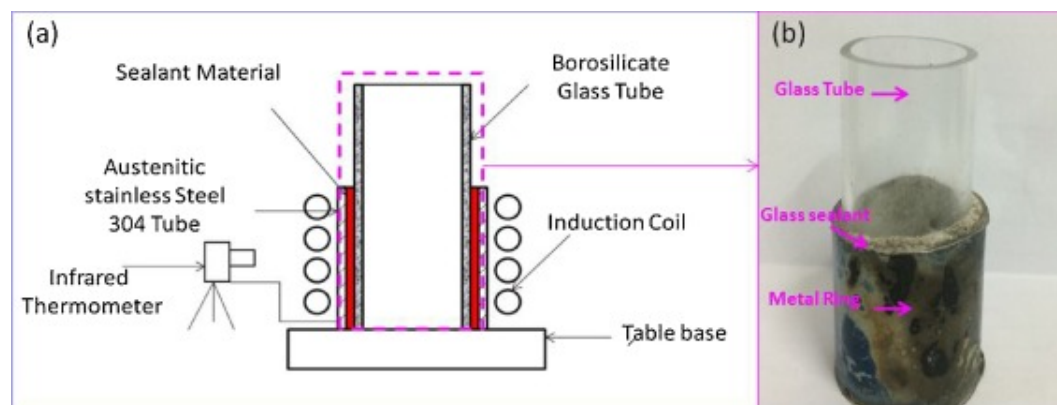
In order to study the diffusion of molten glass sealant over austenitic stainless steel SS304 metal specimen, the glass sealant pallet was melted over metal substrate (pre-oxidized at 700 °C for 3 min) and having cross-sectional area of 25 mm × 25 mm with 1.5 mm thickness. In order to see the morphology and presence of oxide at glass-metal interface of the above mentioned specimen a 10 mm × 10 mm × 1.5 mm square specimen was cut from the above mentioned specimen and then mounted in a sample holder filled with resin. The sample was then polished using specimen cloth polishing machine to obtained flat surface. The polished specimen was then analysed using “Bruker Axis Automated

Powder X Ray Diffractometer". The X-ray diffraction pattern obtained was then analysed with the help of software (ICDD) "International Centre for Diffraction Data". The specimens were then examined under Scanning Electron Microscope of Zeiss make. The composition of element present at interface was obtained with the help of EDAX "Energy Dispersive X-rays Spectroscopy". The X-ray dot mapping was used to see the distribution of different element distribution over the interface in these specimens.

2.4. Glass-metal joint fabrication

In order to have reliable and automatic fabrication process the induction heating set-up was used.

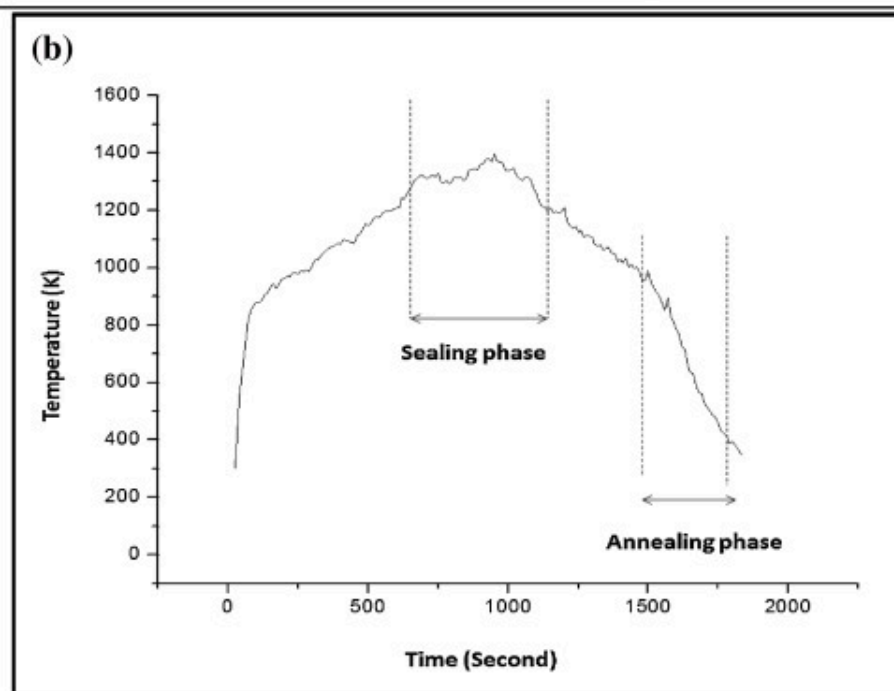
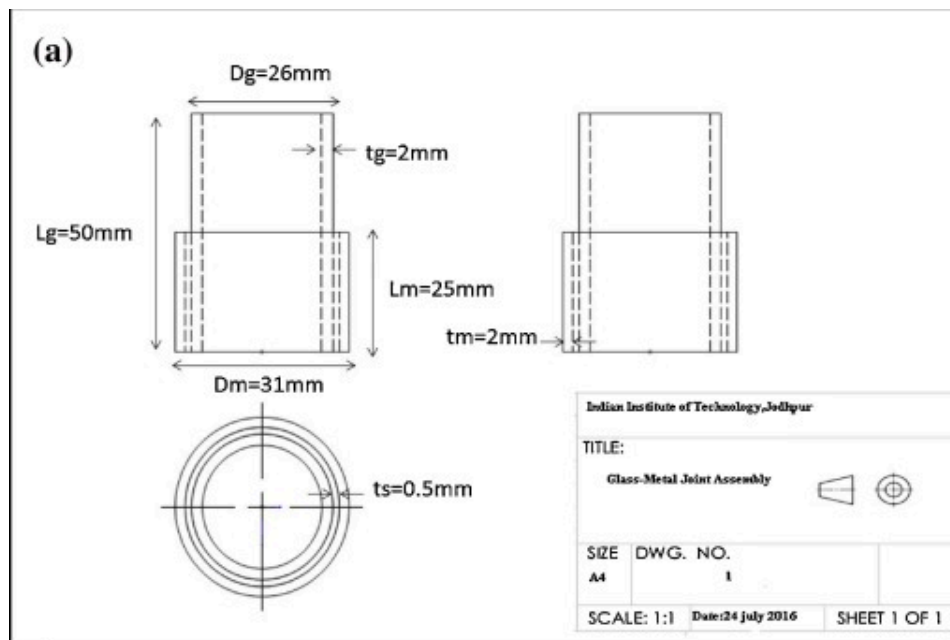
The schematic diagram of glass-metal joint assembly placed over induction heating setup is shown in Fig. 4(a). For the fabrication of compression glass-metal joint, glass tube along with metal was initially washed with soap solution and then dipped into acetone for 2 min.



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Fig. 4. (a) Schematic of experimental set-up. (b) Fabricated glass-metal compression joint.



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Fig. 5. (a) Dimensions of glass-metal assembly. (b) Heating curve of glass-metal joint.

Metal ring was first pre-oxidized to 700 °C for 3 min to produce the oxide layer on metal ring. Glass sealant material was initially in the form of powder, then 3% (wt) polyvinyl alcohol (PVA) binder was added to glass powder slowly and mixed in mortal pastel to bind the glass powder. The solid wet glass sealant is then filled between the space of glass tube and the metal ring and then the assembly is baked in oven for 120 °C to evaporate the binder from the glass sealant powder.

The chemical composition of the borosilicate glass tube, SS304 metal specimen and glass sealant used for present work are shown in [Table 1](#), [Table 2](#). The dimension of glass to metal joint assembly used for present study is shown in [Fig. 5\(a\)](#). Glass-metal joint assembly was then heated slowly in induction heating setup, until glass sealant melts in between the space of glass tube and metal ring. Finally the assembly was cooled slowly to release the thermal stresses. The variation in heating temperature with time for the glass-metal joint was recorded using infrared thermometer as shown in [Fig. 5\(b\)](#).

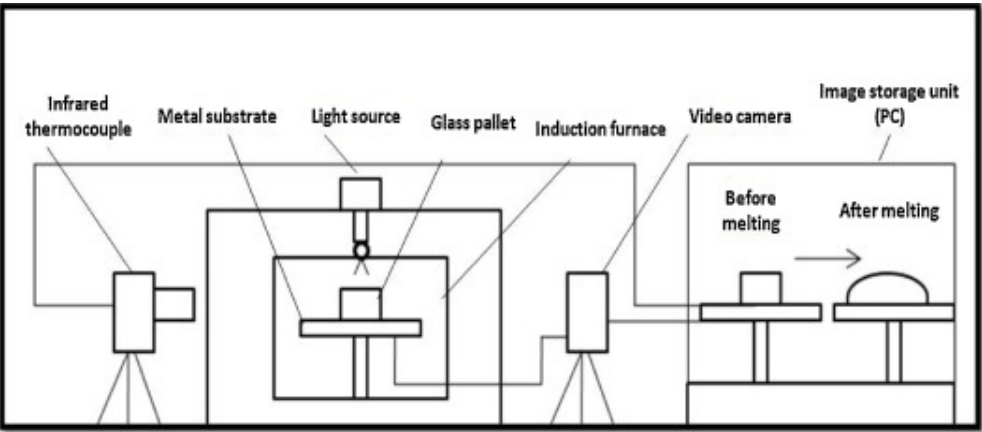
Table 2. Chemical Composition of glass, metal and glass tube.

Material	Composition (Wt%)							
Austenitic Stainless Steel 304 Ring SS304)	C	Mn	P	S	Cr	Ni	N	Fe
	0.08	2	0.045	0.030	18	12	0.10	Balance
Borosilicate Glass Tube	SiO ₂	B ₂ O ₃	Na ₂ O	Al ₂ O ₃				
	80	13	4	3				
Sealant material	SiO ₂	B ₂ O ₃	Na ₂ O	Al ₂ O ₃				

Material	Composition (Wt%)			
	40	54	4	2

2.5. In-situ wetting behavior of glass sealant

High-temperature image capturing system allows direct in-situ observation of melting and softening behavior of glass over metal substrate [17]. The change in shape of glass sealant pallet from room temperature to 980 °C was monitored using digital camera along with thermo couple in an induction heating system furnace. The schematic of image capturing system is shown Fig. 6, it consist of induction heating set-up, image capturing digital camera, computer, focusing light and infrared thermometer.



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Fig. 6. A schematic of experimental setup for in-situ wettability measurement.

The experimental set-up uses an induction coil, metal strip of SS 304 having dimension 25 mm × 62.5 mm × 1.5 mm and glass sealant pallet of 8 mm diameter and 10 mm height. The glass sealant pallet was prepared from sealant glass composition as shown in Table 2. Glass sealant powder is then mixed in mortar pestle by adding 3% (by weight) polyvinyl

alcohol (PVA) in water and glass sealant powder mix was then filled into the pallet forming die. The die along with filled powder mix is then pressed in press having 10 ton capacity. A pressure of 100 kg/cm^2 is then applied on powder pressing die for 1 min. The pressed powder mix in the form of glass sealant pallet is then removed from the die and baked at 120°C in oven to remove (PVA) from the glass sealant pallet. The glass sealant pallet was then placed over pre-oxidized SS304 substrate (pre-oxidised at 700°C for 3 min) and complete assembly was placed in induction heating system, in order to provide localized heat to metal substrate. Infrared thermometer was installed to measure the temperature of glass-metal assembly. A reference object having known dimension is marked in front screen in order to calculate and correlate image pixel size measured with respect to reference object. The melting behavior of glass sealant pallet at different stages from room temperature to 980°C was recorded using digital camera. The captured video image was then converted into still image frames. The captured video before melting was converted into still image and measurement of the initial height of the glass sealant pallet in terms of pixel was carried out.

The actual height of glass sealant pallet was 10 mm and the number of pixel obtained in still image captured were 230 pixel.

1 mm of length = $230/10$ pixel

Similarly, 1 pixel represents $10/230$ mm

This measurement was used to calculate the height (h) and diameter (d) at different stages during melting and solidification to obtain the value of contact angle using formula

$$\theta/2 = \tan^{-1}(h/d) \quad (1)$$

Where, d = diameter of the glass sealant pallet

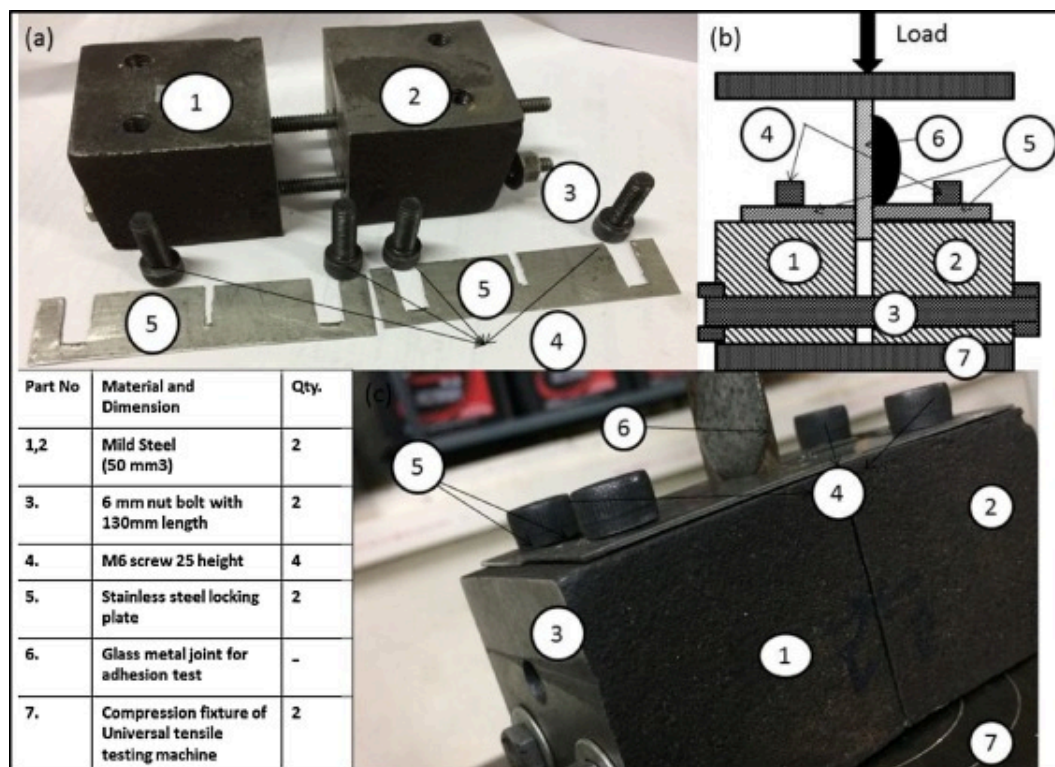
h = height of glass sealant pallet

θ = contact angle measured

2.6. Adhesion test

Adhesion test between glass and metal was performed under compression loading condition.

For glass-metal adhesion test glass sealant pallet is melted over pre-oxidized metal specimen and adhesion between glass sealant pallet and metal specimen were measured using specially designed fixture shown in [Fig. 7\(a\)](#) where load presses only on the metal part (dimension: 25 mm × 25 mm × 1.5 mm) which is fixed by locking the stainless steel plate as shown in [Fig. 7\(a–c\)](#). Compression testing was performed by placing specially designed fixture in UTM “Universal Tensile Testing Machine of Lloyd Instruments, Ez-50 model”. The load where glass detaches from the surface of metal piece is recorded in computer attached with the UTM. The dimensions and material used for fabricating this fixture is shown in [Fig. 7](#).



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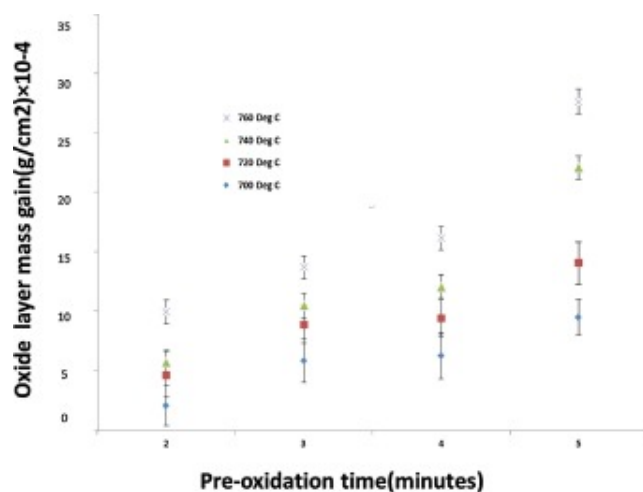
Fig. 7. Specimen mounting holder for adhesion test.

3. Results and discussion

The unmatched glass-metal joint was fabricated using compression geometry as shown in Fig. 5(a). The glass-metal joint was fabricated on the basis of pre-oxidation study of the substrate and the interfacial study involving glass sealant pallet and SS304 metal substrate. The temperature during the sealing was recorded using the infrared thermometer as shown in Fig. 5(b).

3.1. Oxide layer mass gain and oxide layer thickness

The oxide layer mass gain of metal specimen measured at different pre-oxidation time and pre-oxidation temperature is shown in Fig. 8. The oxide layer mass gain of SS304 metal specimen increases with increase in oxidation time (2–5 min) at 700 °C. As pre-oxidation temperature increases above 700 °C the oxide layer mass gain increased. This is due to the increased formation of stable oxides FeO and Fe₂O₃ due to internal oxidation in the temperature range (700 °C–760 °C) which remain bonded to the metal and are increasingly deposited as the pre-oxidation temperature increases and pre-oxidation time increases. The growth of oxide layer on the metal surface also takes place due to the decomposition of H₂O. The dissociative oxygen transports to the metal interface and convert into oxygen ion. The Fe atom present into the metal specimen change into ionic form and electron which diffuses into the surface of metal.



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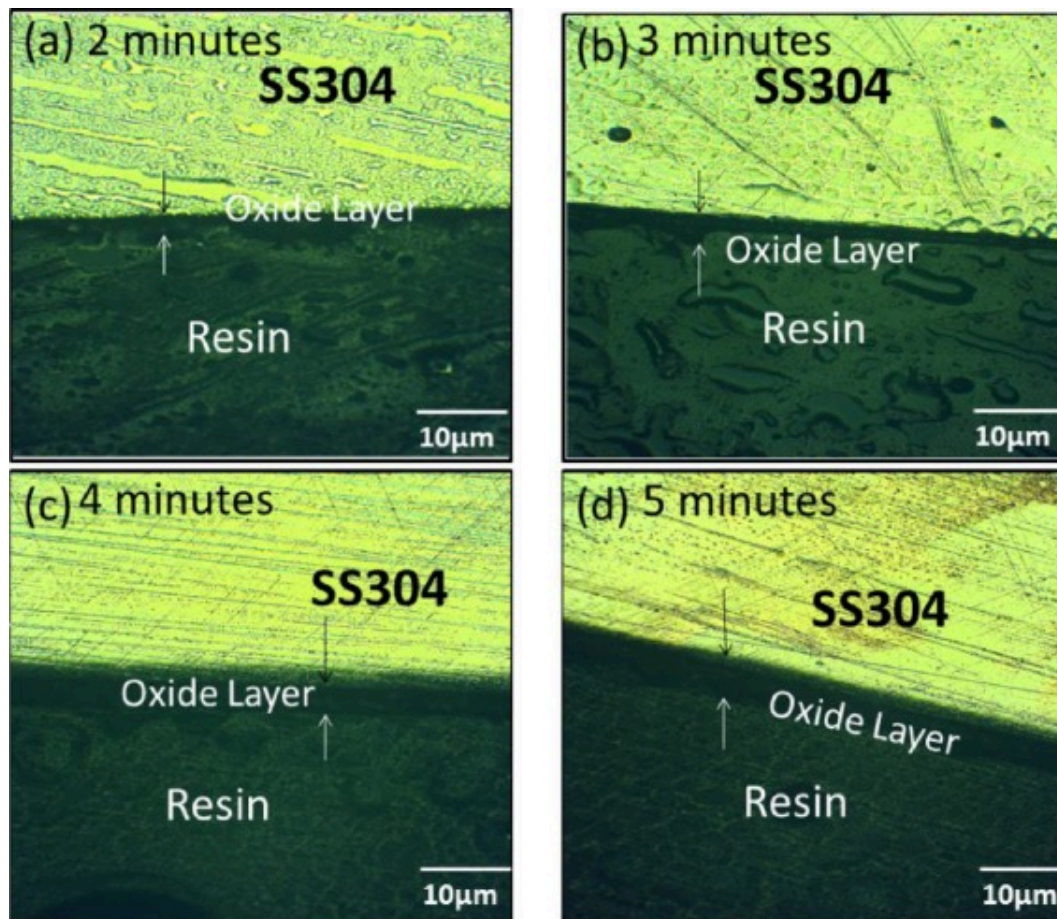
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Fig. 8. Oxide layer mass gain at different pre-oxidation temperature and different pre-oxidation time.

The optimum oxide layer thickness on metal specimen varies with type of metal. The oxide layer mass gain for austenitic stainless steel substrate was observed to be in range

($2.09 \times 10^{-4} \text{ g/cm}^2$ – $9.5 \times 10^{-4} \text{ g/cm}^2$) at 700 °C pre-oxidation temperature and for pre-oxidation time intervals 2–5 min.

The optimum oxide layer mass gain ($5.8666 \times 10^{-4} \text{ g/cm}^2$) was achieved corresponding to pre-oxidation temperature 700 °C for and pre-oxidation time interval of 3 minutes. At this condition the maximum spreading (spreading area) of glass sealant was observed. As pre-oxidation temperature increases above 700 °C the surface degradation (flake surface) of metal specimen was observed which reduces the wettability of glass sealant over the metal substrate. The 700 °C temperature was considered to be the most suitable pre-oxidation temperature where maximum wettability of glass with SS304 metal was observed and similar result was also reported for kovar alloy by other researchers [[11], [14]]. Hence, further study was carried out for the measurement of oxide layer thickness at 700 °C by varying pre-oxidation time (2–5 minutes) to obtain oxide layer thickness in range of (69.87–132.5 µm). The oxide layer thickness formed on the surface of metal at 700 °C at different pre-oxidation time interval for (2–5 minutes) was measured using an optical microscopy as shown in Fig. 9(a–d). The oxide layer thickness increases as pre-oxidation time increases as shown in Fig. 9(a–d).



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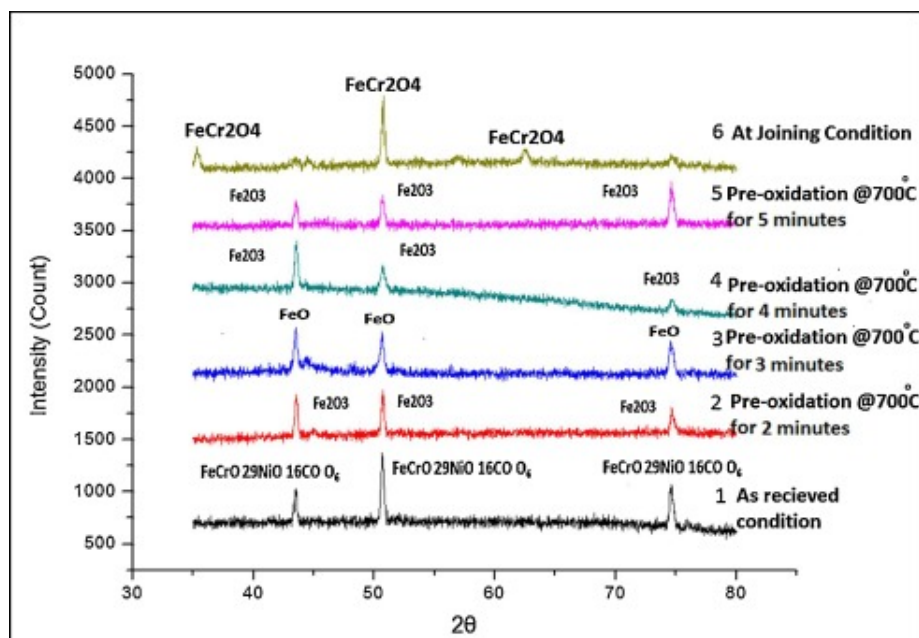
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Fig. 9. Oxide layer thickness (μm) at pre-oxidation temperature 700 °C at different pre-oxidation time interval.

3.2. Types of oxide formed after pre-oxidation

The XRD analysis of SS304 alloy specimen in as received condition, pre-oxidation condition and at joining condition is shown in [Fig. 10](#). The as received SS304 metal specimen consists of chromium, iron, nickel and carbon phases. As metal is pre-oxidized

at 700 °C for 3 min, it was observed that FeO phase was formed in the alloy specimen as shown in Fig. 10. Similar types of oxide formation were also reported for kovar alloy after pre-oxidation by D.W. Luo and Z.S. Shen [10]. At pre-oxidation temperature 700 °C for 3 min, the Fe to O ratio was observed near to 1 as shown in Table 3. The Fe to O ratio confirms the presence of FeO (wustite) type of oxide phase having closed cubic configuration, formed on the surface of metal specimen because of oxidation. The similar types of oxide were also observed in XRD analysis. The growth of FeO oxide phase on the metal surface takes place due to the decomposition of H₂O. The dissociative oxygen transports to the metal interface and converts into oxygen ion. The Fe atom present in the metal specimen converts into ferrous ion and electron which diffuse into the surface of metal using reaction given in equation:



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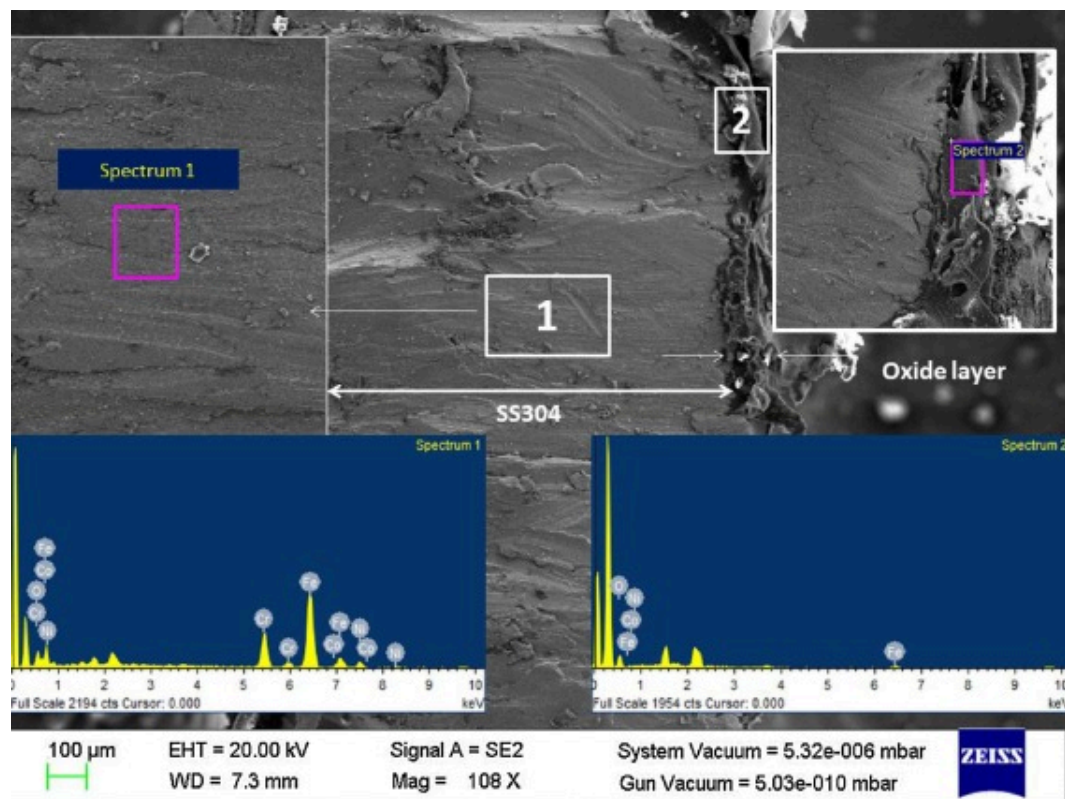
Fig. 10. XRD pattern of SS304 metal substrate during, as received, pre-oxidation and joining condition.

Table 3. EDAX elemental composition on SS304 specimen after pre-oxidation.

	Weight %				
	Fe	Co	Ni	Cr	O
Spectrum 1	66.67	0.75	5.32	18.52	6.44
Spectrum 2	46.97	4.84	1.27	–	46.92

The glass pallet during glass-metal joint fabrication process melts over the pre-oxidized metal specimen and reacts with (FeO) oxide phase on metal specimen and forms adhesion with glass. The rapid diffusion of Fe^{2+} which is a glass former and highly soluble leads to the formation of diffusion gradient which is responsible for the strong adhesion at the glass-metal joint interface. At 700 °C pre-oxidation temperature and corresponding to 2–5 min pre-oxidation time the Fe_2O_3 oxide phase was observed.

The FeO type of oxide was reported as the most suitable form of oxide for adhesion whereas use of Fe_2O_3 phase was not recommended because of its loose structure. At the joining temperature 980 °C, FeCr_2O_4 phase dominates as observed in the XRD pattern as shown in [Fig.10](#). The scanning electron microscope images of oxide layer deposited on the surface of metal substrate after pre-oxidation at 700 °C for 3 min is as shown in [Fig. 11](#). The elemental compositions obtained with energy dispersive x-ray spectroscopy measured for region marked 1 and 2 in [Fig. 11](#) is shown in [Table 3](#) as spectrum 1 and spectrum 2.



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Fig. 11. Oxide layer deposited on the SS304 metal specimen after pre-oxidation.

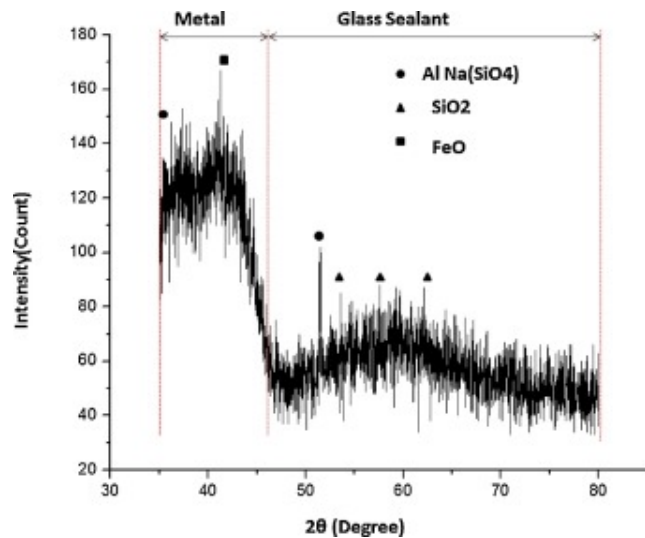
The elemental compositions measured for marked region 1 show the percentage of elements Fe, Cr, Ni, Co, O which is nearly close to the elemental composition for SS 304 as shown in Table 2 which confirms the presence of SS 304 in the marked region 1. The region 2 shows the percentage of oxide scale where Fe to O ratio was observed near to 1.001 which confirms the presence of FeO phase at the oxide layer. The phase formed and elemental composition of the phase at surface of metal after pre-oxidation of metal specimen is shown in Table 4. Elemental composition obtained with EDAX along with oxide layer mass gain, oxide layer thickness, Fe to O ratio and oxide phase at different pre-

oxidation time is shown in Table 3. The FeO phase was obtained after pre-oxidation at 700 °C for 3 min where maximum adhesion was observed whereas Fe₂O₃ oxides were present at 2–5 min.

Table 4. Oxide layer mass gain and oxide layer thickness at different pre-oxidation time intervals.

At 700 °C pre-oxidation	Oxide layer mass gain(g/cm ²)	Oxide layer thickness(μm)	Elemental Composition				Fe/O Ratio	
			Fe	Co	Ni	O	Oxide Phase	
2 min	2.09358×10^{-04}	69.87	35.87	3.74	4.65	55.74	0.64	Fe ₂ O ₃
3 min	5.86667×10^{-04}	87.57	46.97	4.84	1.27	46.92	1.001	FeO
4 min	6.31795×10^{-04}	116.64	38.34	0.25	5.1	56.31	0.67	Fe ₂ O ₃
5 min	9.54584×10^{-4}	132.95	36.56	2.47	6.1	54.87	0.66	Fe ₂ O ₃

The identification of these phases was confirmed on the basis of Fe to O ratio obtained with energy dispersive X-ray spectroscopy as shown in Table 3 and XRD analysis as shown in Fig. 10. During the XRD analysis FeO, AlNa(SiO₄) and SiO₂ types of phases were present at glass- metal interface as shown in Fig. 12. Similar types of oxides were also reported by Kaur B et al. [16].

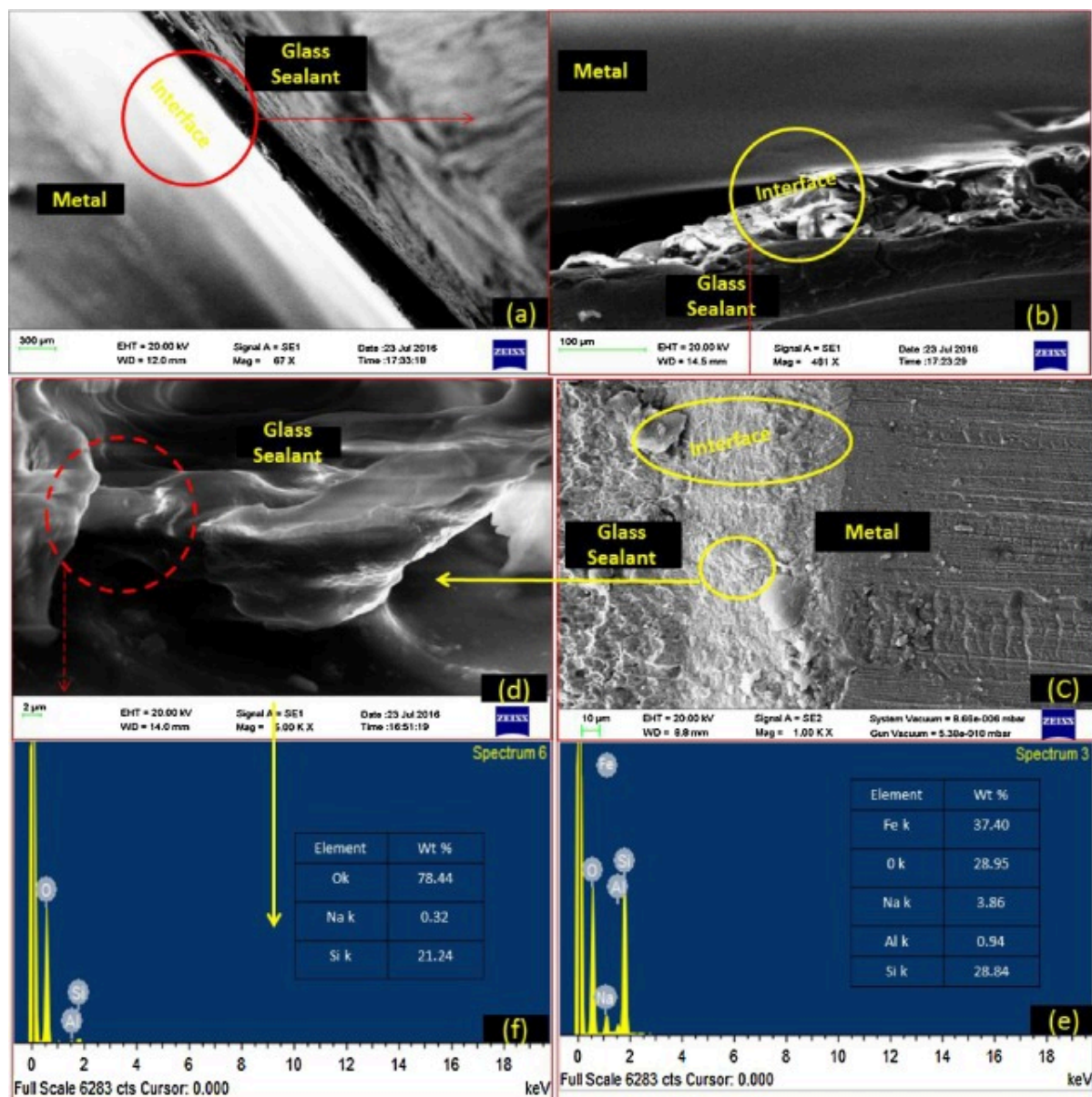


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Fig. 12. XRD pattern of glass sealant-metal interface.

The presence of FeO, AlNa(SiO₄) and SiO₂ types of phases at glass sealant-metal interface confirmed the presence of both glass and metal as shown in Fig. 12. The FeO phase confirms the metal oxide part whereas AlNa(SiO₄) and SiO₂ confirms the presence of glass phase containing Na₂O and Al₂O₃. The morphology of interface at different magnification is as shown in Fig. 13(a,b) and a crack free joint is obtained. The morphology of glass region at higher magnification 6000X on scale of 2 μm is shown in Fig. 13(c).



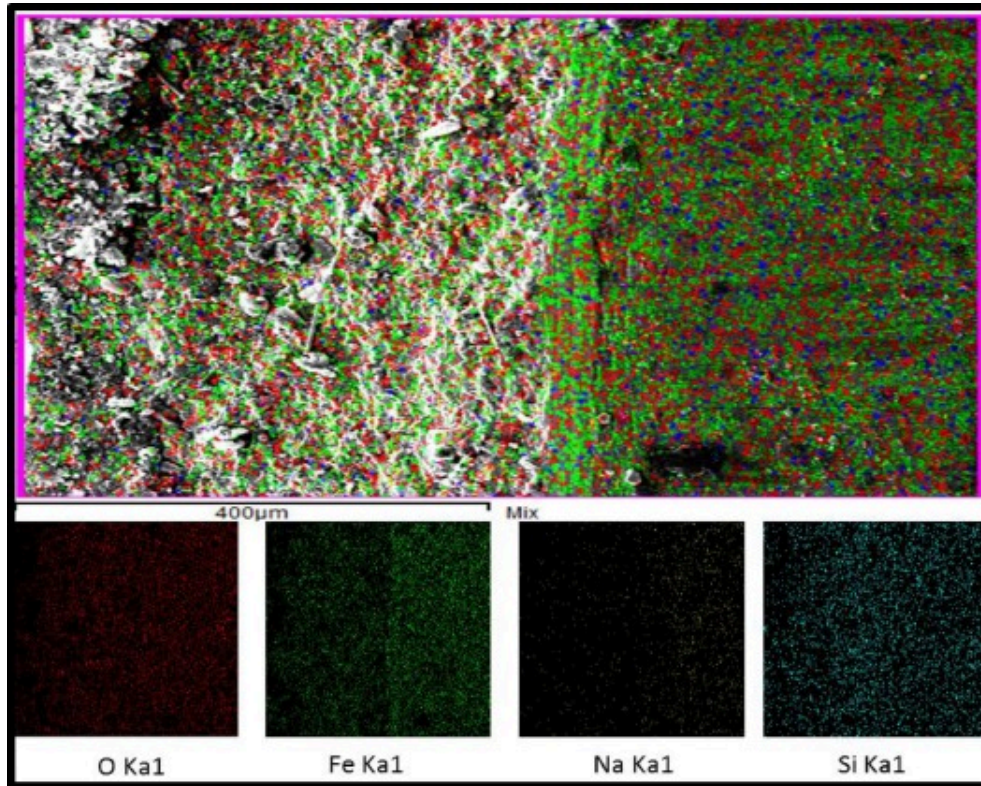
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Fig. 13. (a) SEM image with scale 300 μm, (b) SEM image with scale 100 μm, (c) SEM image with scale 10 μm, (d) SEM image with scale 2 μm, (e) EDAX measurement taken

corresponding to SEM image with scale 2 μm , (f) EDAX measurement taken corresponding to SEM image with scale 10 μm .

The interface of glass sealant-metal joint was analysed using EDAX (Energy Dispersive X-ray Spectroscopy) for determining different elements such as Fe(37.40 wt%), O(28.96 wt%), Na(3.86 wt%), Al(0.99 wt%) and Si(26.84 wt%). The distribution of elements (Fe, Na, O and Si) at the glass sealant-metal interface as shown in Fig. 14 with X-ray dot mapping confirms the presence of these elements at the interface. The different color dot describes the distribution of different element on the interface region of the glass-metal joint. In order to see the morphology of glass-metal interface at higher magnification it was further observed under scanning electron microscope at magnification of 1000X on a scale of 10 μm as shown in Fig. 13(d). The EDAX of region marked at glass-metal interface was observed in Fig. 13(f). At glass sealant-metal interface marked in region Fig. 13(d) the elements Fe, Si, Na, Al and O were present which confirm the presence of glass and metal oxides. The presence of glass was confirmed with the presence of Si, O and Al at magnification of 6000X with scale of 2 μm with the help of EDAX analysis as shown in Fig. 13(e).



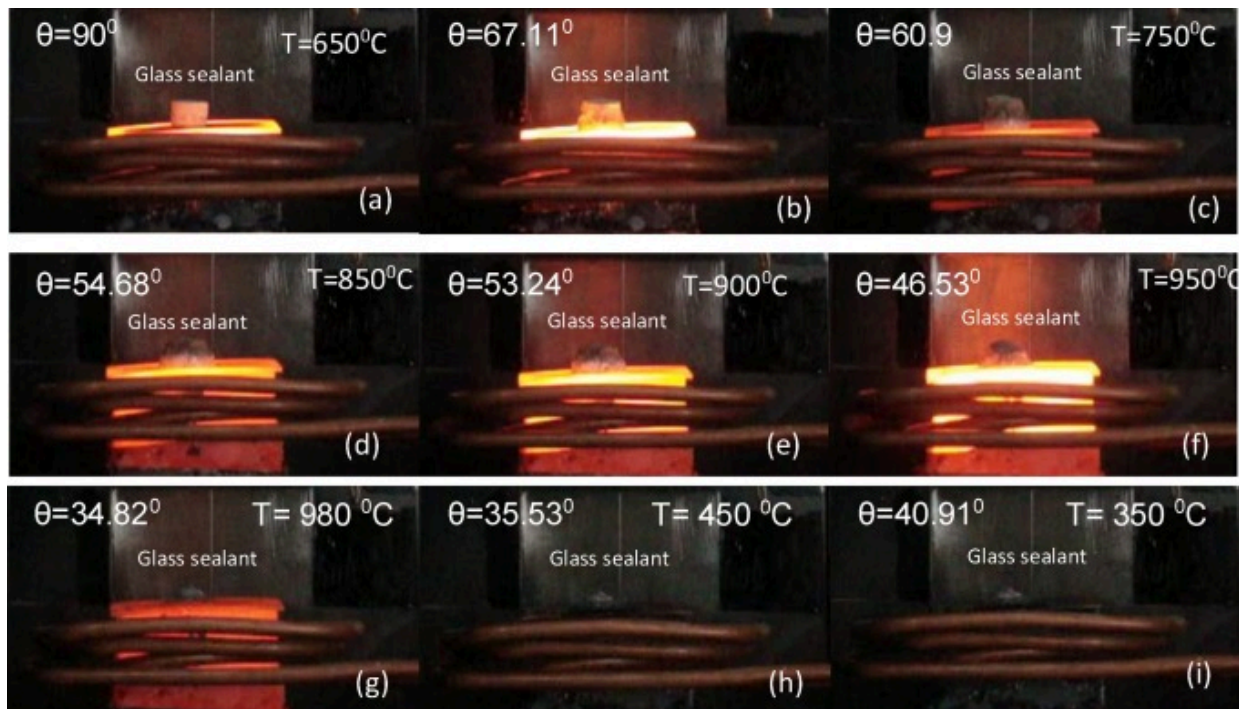
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Fig. 14. X ray dot mapping at the interface of glass sealant-metal joint.

3.3. In-situ wetting behavior of glass sealant

The in-situ wetting behavior of glass sealant over SS304 metal substrate studied at different temperatures is shown in [Fig.15](#).



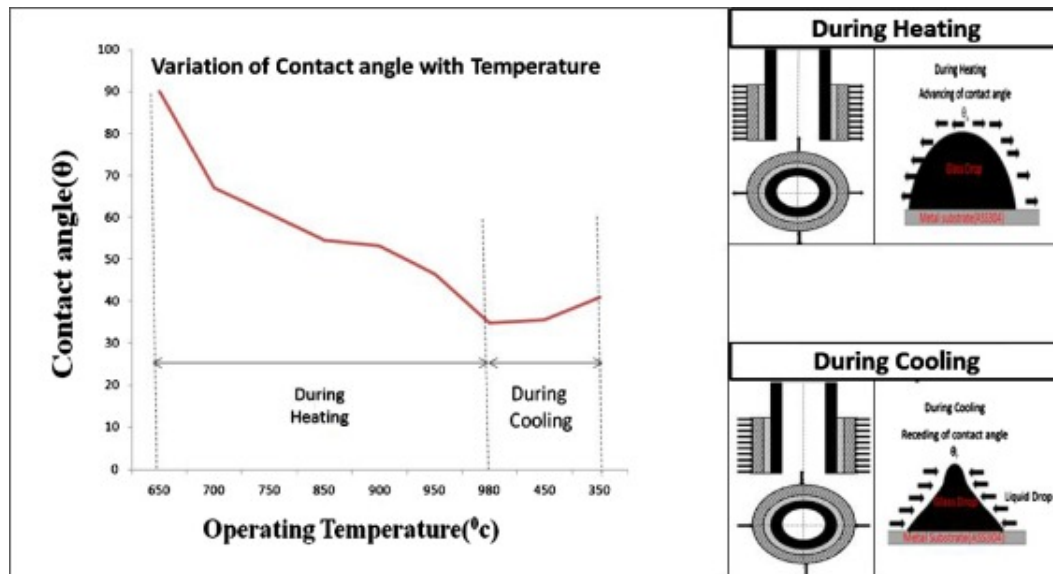
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Fig. 15. In-Situ Wetting Behavior of Glass Sealant over metal substrate at different temperature.

The contact angle at different temperature was measured using Digimizer (Image analysis software). During the in-situ wettability analysis of glass sealant over metal specimen, contact angle decreases with increasing temperature as shown in Fig. 15(a–g) and as glass cools from melting temperature to room temperature, contact angle starts increasing during cooling shown in Fig. 15(g–i). The variation of contact angle of glass with respect to temperature is shown in Fig. 16. The initial contact angle of glass sealant pallet measured was 90° and as the temperature increases from room temperature to 980°C the value of contact angle starts decreasing from 90° to 34.82°. As glass sealant pallet cools from 980°C to 340°C the contact angle value starts increasing from 34.82° to 40.91°. The

increase in contact angle is due to the shrinking of the glass. During glass-metal joint fabrication as shown in Fig. 16 it is observed that the glass expands during heating due to the expansion of the metal ring in outward radial direction. Whereas as the assembly cools from 980 °C to room temperature the metal ring starts producing radial compressive force in glass sealant and glass tube, hence the contact angle of glass sealant pallet starts increasing which confirms that the glass sealant has been subjected to compressive loading during cooling. The resulting compressive force in the glass sealant and the glass tube allows the joint to resist high mechanical stresses.



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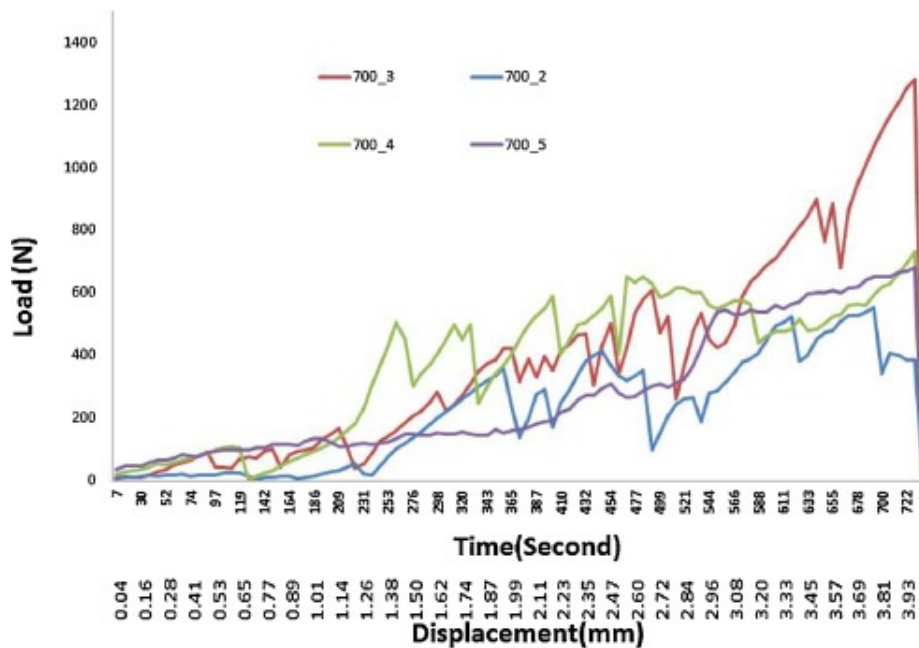
Fig. 16. Variation of contact angle with temperature.

3.4. Adhesion test

The adhesion between the glass metal joining process depends upon the wettability and pre-oxidation of metal specimen. In the process of adhesion process glass diffuses into

the pre-oxidized metal specimen where glass reacts with the oxide present at the surface of metal specimen.

The adhesion of the glass sealant-metal joint was evaluated with change in wettability. The wettability was measured by evaluating the spreading area of the glass sealant pallet. The result of adhesion test for glass sealant- metal joint along is shown in Fig. 17. The mechanical load versus displacement behavior under applied compressive load was recorded using Universal Tensile Testing Machine as shown in Fig. 17. The maximum compressive load was observed for glass sealant –metal specimen pre-oxidised at 700 °C pre-oxidation temperature and for 3 min pre-oxidation time. The maximum load bearing capacity of the glass sealant-metal joint ensured its maximum adhesion.

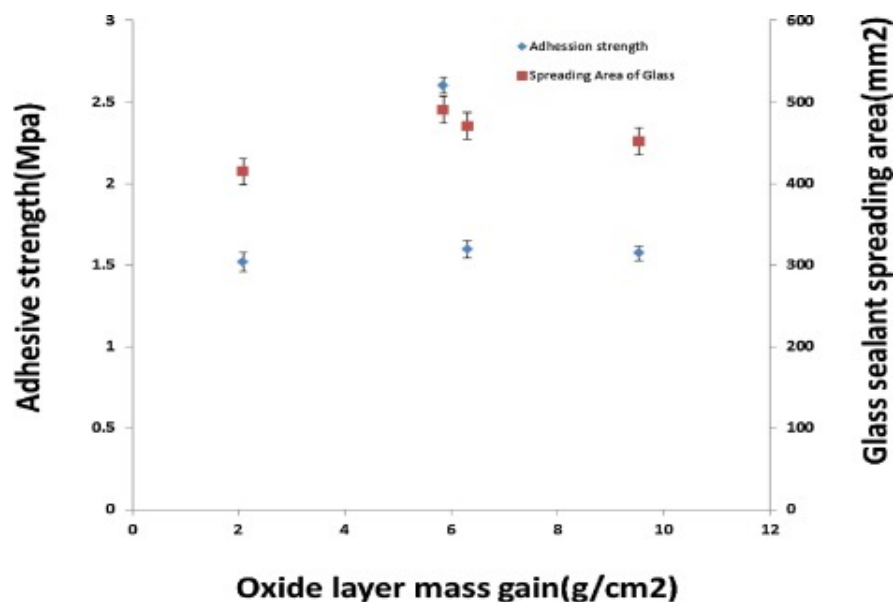


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Fig. 17. Load versus extension variation of glass-metal joint during adhesion test.

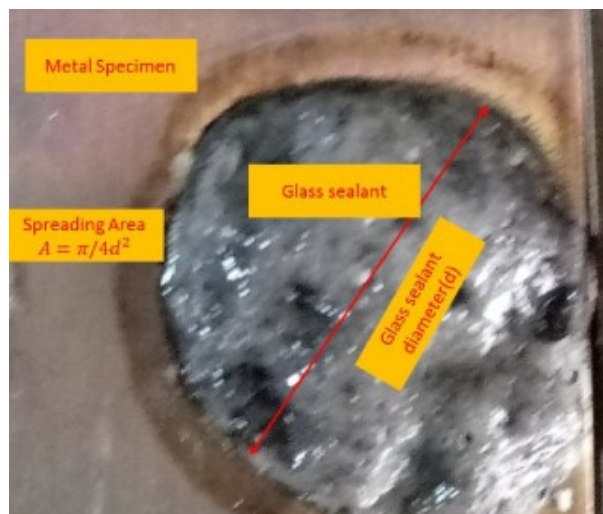
It was observed that the maximum compressive load for the glass sealant- metal joint was obtained for glass sealant pallet with maximum spreading area (490 mm^2) shown in Fig. 18. The spreading area of glass sealant pallet was evaluated through the visual examination of each glass sealant pallet shown in Fig. 19. The experimental result of spreading area and adhesion strength with change in oxide layer mass gain at 700°C pre-oxidation temperature for different time intervals is shown in Fig. 18.



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Fig. 18. Adhesion strength and spreading area of glass sealant with change in oxide layer mass gain observed at 700°C pre-oxidation temperature and pre-oxidation time of 2–5 min.



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Fig. 19. Estimation of spreading area of glass sealant over metal.

Energy Dispersive X-Ray Spectroscopy measurements at the interface of glass-metal joint confirm the presence of elements: Fe(37.40 wt%), O(28.96 wt%), Na(3.86 wt%), Al(0.99wt%) and Si(26.84 wt%). The adhesion between glass sealant -metal joint required a continuous transition from the metallic structure of metal to ionic stage of glass sealant as shown in X-ray dot mapping of the joint interface. The adhesion strength of joint depends upon the wettability of the glass sealant over metal specimen. The maximum adhesion strength of glass sealant-metal joint is 2.57 N/mm^2 (MPa) corresponding to 700°C pre-oxidation temperature and pre-oxidation time of 3 min.

4. Conclusion

From the present study the following major conclusions are drawn.

- The optimum oxide layer mass gain for the joining of SS304 metal specimen and borosilicate glass was observed to be $5.8666 \times 10^{-04} \text{ g/cm}^2$. The optimum

pre-oxidation temperature and pre-oxidation time was 700 °C and 3 min respectively. FeO phase was the major phase obtained from the pre-oxidation of metal specimen at 700 °C for 3 min. For pre-oxidation time of 2–5 min the Fe₂O₃ was observed as the major phase present and adhesion strength of joint was observed to be lower as compared to FeO phase. The lower adhesion strength corresponding to Fe₂O₃ phase was attributed to the loose structure of Fe₂O₃ as compared to FeO. The rapid diffusion of Fe²⁺ which is a glass former and highly soluble leads to the formation of diffusion gradient which is responsible for the strong adhesion at the glass sealant-metal joint interface.

- The maximum adhesion strength of 2.57 N/mm²(MPa) for glass sealant-metal joint was observed at 700 °C with 3 min pre-oxidation time, where the oxide layer mass gain was observed to be (5.8666×10^{-04} g/cm²) and optimum oxide layer thickness measuring 87.57 μm was obtained. The maximum wettability was obtained at 700 °C with 3 min pre-oxidation time measured by measuring the spreading area (490 mm²) of the glass sealant pallet.
- At the glass-metal interface both glass and metal phase exist. AlNa(SiO₄), (SiO₂) and FeO phases were identified at the glass-metal interface region during using XRD measurement. The presence of FeO formed after the pre-oxidation of metal substrate specimen bonds with glass sealant. The presence of FeO, ((SiO₂), AlNa(SiO₄)) phases at the interface of glass sealant-metal joint result in strong adhesion between glass sealant and the metal.
- From the scanning electron microscopy examination the quality of joint examined was found to be crack free. The elemental composition measured using EDAX at the glass sealant-metal interface show the elements: Fe(37.40 wt%), O(28.96 wt%), Na(3.86 wt%), Al(0.99 wt%) and Si(26.84 wt%). The presence of these elements at glass sealant-metal interface confirmed

the bonding of glass sealant and metal at the interface. The distribution of elements (Fe, Na, O and Si) at the glass sealant-metal interface as analysed with X-ray dot mapping confirms the distributed presence of these elements at the interface. The adhesion between glass sealant and metal is due to a continuous transition from the metallic structure of metal to ionic stage of glass sealant in the glass-metal joint.

- The wettability of glass sealant over metal substrate increased with increasing temperature. As glass-metal assembly is cooled from melting temperature to room temperature, the contact angle increases due to the shrinkage of glass sealant pallet, which confirms that glass sealant produces compressive force on glass tube, as it is cooled from sealing temperature.

Acknowledgement

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...Joshi and Chhibber (2016) studied the wettability of borosilicate glass and found that pre-oxidation of austenitic stainless steel at 700 °C for 3 min pre-oxidation time increases the wettability of borosilicate glass. Joshi and Chhibber (2018) also studied the pre-oxidation behavior of austenitic stainless steel SS304 by varying the pre-oxidation time (2,3,4,5) and observed that FeO types of oxide phase was obtained at pre-oxidation temperature 700 °C for 3 min whereas Fe₂O₃ types of oxide phase was observed corresponding to pre-oxidation temperature 700 °C for 2, 4 and 5 min time interval. The entire glass sealant specimen was heated in the furnace to 980 °C temperature from room temperature....

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